Ex NO:8

Implement Packet Sniffing using Raw Socket in python

Aim:

To develop a basic packet sniffer using python that captures network packets and display their IP header information such as source IP, destination IP, protocol, TTL and packet length.

Procedure:

1. Install python and add it to the System PATH.
2. Create a project folder and file named Sniffer.py
3. write a program into file
4. Open command prompt as administrator
5. Navigate to the project folder
6. Run the program
7. Observe the output

Code:

```
import socket
import struct

sniffer = socket.socket(
        socket.AF_INET,
        socket.SOCK_RAW,
        socket.IPPROTO_IP)
host = socket.gethostbyname(socket.
gethostname())
sniffer.bind((host,0))
sniffer.setsockopt(
```

o/p:

packet sniffer started:

Version: 4

Header length: 20

TTL: 128

protocol: 6

Source IP: 192.168.1.10

Destination IP: 142.250.183.14

packet length: 60

Version: 4

Header length: 20

TTL: 64

protocal: 17

source IP: 192.168.1.10

Destination IP: 8.8.8.8

Packet length: 76

stopped.

```
    socket.IPPROTO_IP,
    socket.IP_HDRINCL,1)
sniffer.ioctl(socket.SIO_RCVALL,
socket.RCVALL_ON)
print("Packet Sniffer started...\n")

try:
    while True:
        raw_data,addr= sniffer
                    .recvfrom(65535)
    IP_header = raw_data[0:20]
    iph = struct.unpack(
            "!BBHHHBBH4s4s",
            ip_header)
version = iph[0]>>4
ihl = iph[0] & 0xf
ttl = iph[5]
protocol = iph[6]
source = socket.inet_ntoa(iph[8])
destination = socket.inet_ntoa
(iph[9])
print("-----------")
print("Version:", version)
print("Header Length:", ihl*4)
print("TTL:", ttl)
print("protocol:", protocol)
print("Source IP:", source)
print("Destination IP:", destination)
print("packet length:", len(raw_data))
except KeyboardInterrupt:
    sniffer.ioctl(socket.SIO_RCVALL,
                  socket.RCVALL_OFF)
print("\n stopped")
```

Result: The python program successfully captures network packets and displays their basic IP header information.